

ICT703 – Module 3 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice Iteration and selection*
2. *Practice Boolean logic*

Activity 1: Loops

Without running python, write the outputs of the following loops:

```
for count in range(5):
```

```
    print(count, end = " ")
```

```
for count in range(5):
```

```
    print(count + 1, end = " ")
```

```
for count in range(1, 4):
```

```
    print(count, end = " ")
```

```
for count in range(1, 6, 2):
```

```
    print(count, end = " ")
```

```
for count in range(6, 1, -1):
```

```
    print(count, end = " ")
```

Put the above in python and see if you get the same answer

Activity 2: Loops and strings

Write a program that asks for a string input and then prints each character on a new line.

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Activity 3: Logic 1

Assume that x is 3 and y is 5. Write the values of the following expressions:

- a) $x == y$
- b) $x > y - 3$
- c) $x <= y - 2$
- d) $x == y$ or $x > 2$
- e) $x != 6$ and $y > 10$
- f) $x > 0$ and $x < 100$

Test the above by writing a program and print out the answers of each line.

Activity 4: For to While

Write code for the for loops below using the while statement.

for count in range(100):

 print(count)

for count in range(1, 101):

 print(count)

for count in range(100, 0, -1):

 print(count)

Activity 5: Tax Calculator

How would you change the tax program code created last week to:

- a) Capture errors in input from the user
- b) Display 0 for any negative tax amount
- c) Create a tax calculator that used a progressive tax system instead of a flat tax rate. That is:
 - a. Between 0 and \$19,999 you pay no income tax
 - b. Between \$20,000 and \$49,999 you pay 10% tax on the amount above \$20,000
 - c. Over \$49,999 you pay a flat 20% tax rate on all your earnings